THE IDEA

— A ~20 mm mote of printed electronics on biodegradable substrate — seed-dropped over a fuel break or burn scar.
— It sprouts into a temporary sensor skin for fire season, reports through cluster heads, and dissolves.
— Zero e-waste, no retrieval, no grid to build — sensing that ENDS by design.
— Materials/transient-electronics program — fundable R&D, not a product. Every earlier rung must succeed first.

MATERIALS PATHWAY

COMPONENT	CANDIDATE MATERIALS / APPROACH
Substrate	cellulose paper, silk fibroin, transient polymer
Electronics	printed Zn/MnO_2 battery, printed sensors, organic TFT
Energy	printed PV cell, thermoelectric, RF harvest
Comms	backscatter (no TX) or ultra-low-power LoRa wake-up
Deploy	seed pod form factor — germinates sensor on moisture

RESEARCH TARGETS

TARGET	GOAL
Mass	<1 g per mote (cluster head <5 g)
Cost	<$1/unit at volume (printed, not assembled)
Life	one fire season (~6 months) then degrade
Degradation	complete in <12 months, non-toxic residue
Comms	duty-cycled LoRa or backscatter — rare TX, local clustering
Power	printed battery + PV/TEG harvest — µW average
Sensing	temp/RH/VOC minimum; optical smoke later

WHY THIS MATTERS

— Fire season is seasonal — permanent sensors waste 11 months.
— Degradable motes match the threat's rhythm: deploy, vanish.
— Billions-scale is materials R&D, not drone fleet — mesh math supports it.
— Precedent: DARPA/NSF computational-dust programs prove it's real.

ROADMAP GATES

— Gate 1: university transient-electronics partner (SR&ED eligible).
— Gate 2: lab demo — printed mote senses temp + reports 10 m via cluster head.
— Gate 3: field trial — 100 motes over 1 ha, one season, recoverable data.
— Gate 4: degradable variant — residue confirms <12 month breakdown.
— Gate 5: scale — seed-drop tied to Anchor/Brick infrastructure.

HONEST STATUS

— TRL ~2 — concept + literature. No ArcticChain mote exists yet.
— This sheet is a research brief, not a buildable BOM — deliberately.
— Swarm architecture is the same — only the node shrinks.
— Do NOT promise motes to funders until Gate 2 demo exists.

ARCTICCHAIN — RURAL NODE FLEET

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SCALE **NTS** **1/1**
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DWG NO **AF-D-101**
UNITS mm

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